

The background features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, creating a modern and dynamic visual effect.

Resin Composite Restorative Materials

Dr Marwa Bahaa

Lecturer of Dental Biomaterials

Definitions

Composite :

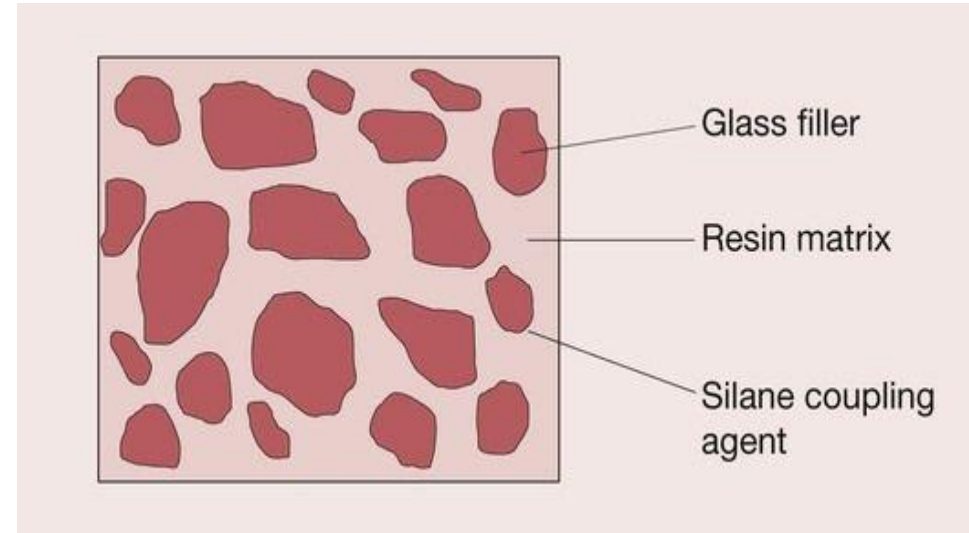
A combination of two or more chemically different materials with a distinct interface between the different components.

Resin Composite:

A resin material used in dentistry as an esthetic restorative material.

Composition

1. Organic matrix.
2. Inorganic fillers.
3. Coupling agent.
4. Initiator-activator system.
5. Polymerization inhibitor.
6. Ultraviolet stabilizer.
7. Pigments.



1. Organic matrix (Dispersion Medium):

Composition :

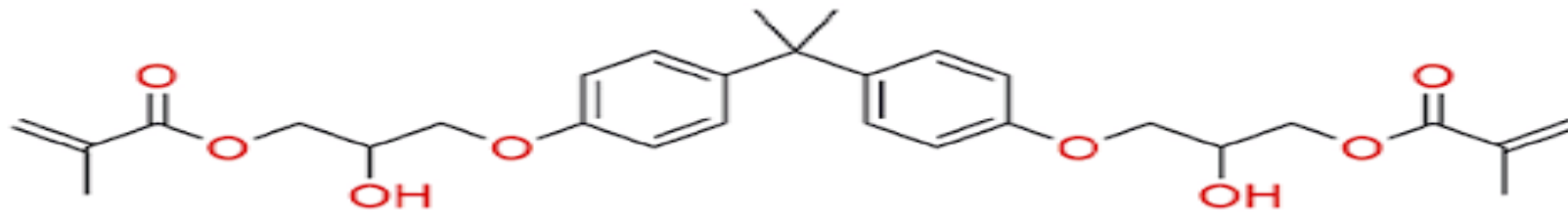
- a) High molecular weight monomer (Oligomer).
- b) Low molecular weight monomer (Diluent).

1. Organic matrix (Dispersion Medium):

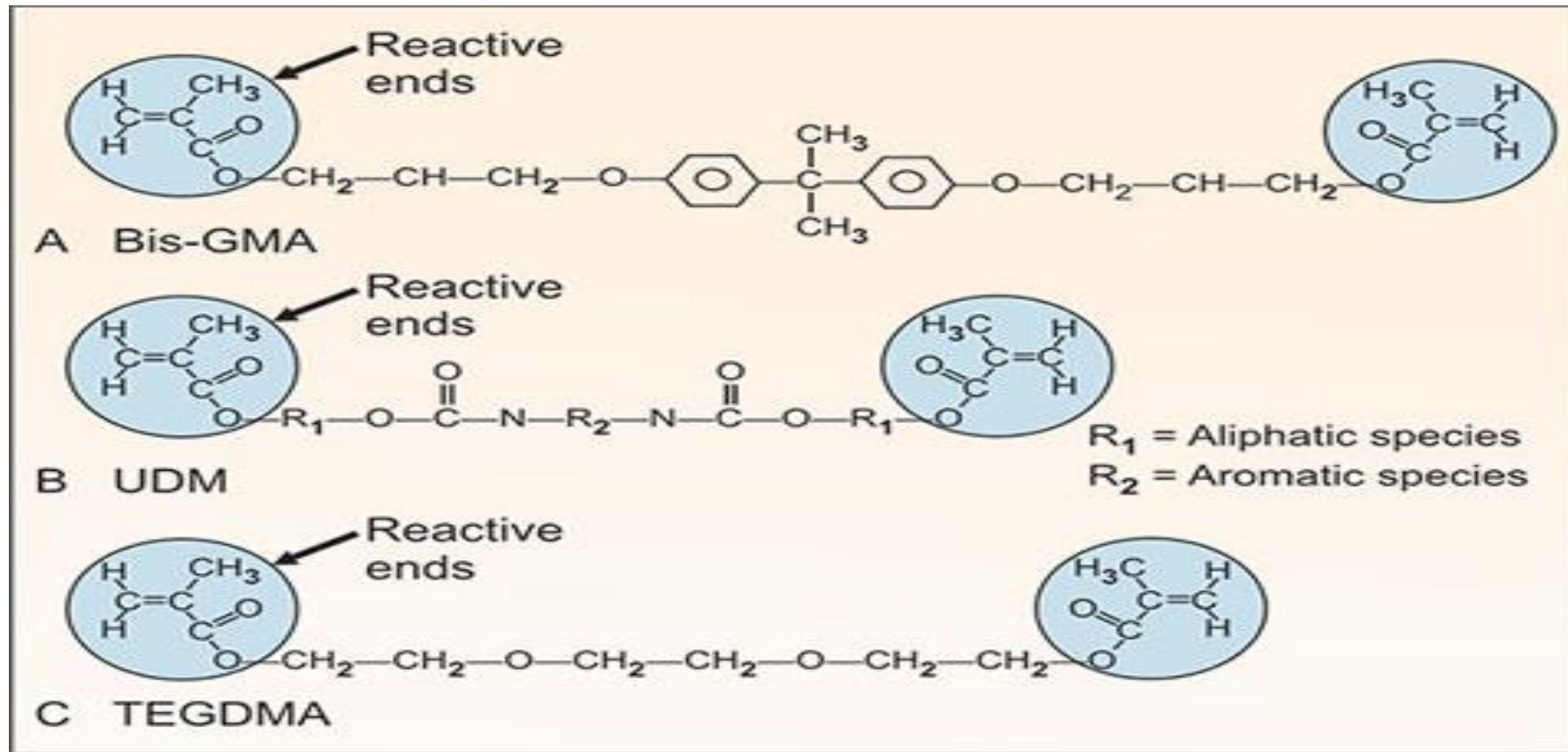
a) High molecular weight monomer (Oligomer):

Types:

1. Bisphenol A-glycidyl methacrylate (Bis-GMA) which is called Bowen's resin.
2. Urethane dimethacrylate (UDMA).
3. Mixture of both of them.



1. Organic matrix (Dispersion Medium):



1. Organic matrix (Dispersion Medium):

a) High molecular weight monomer (Oligomer):

Functions of Oligomers:

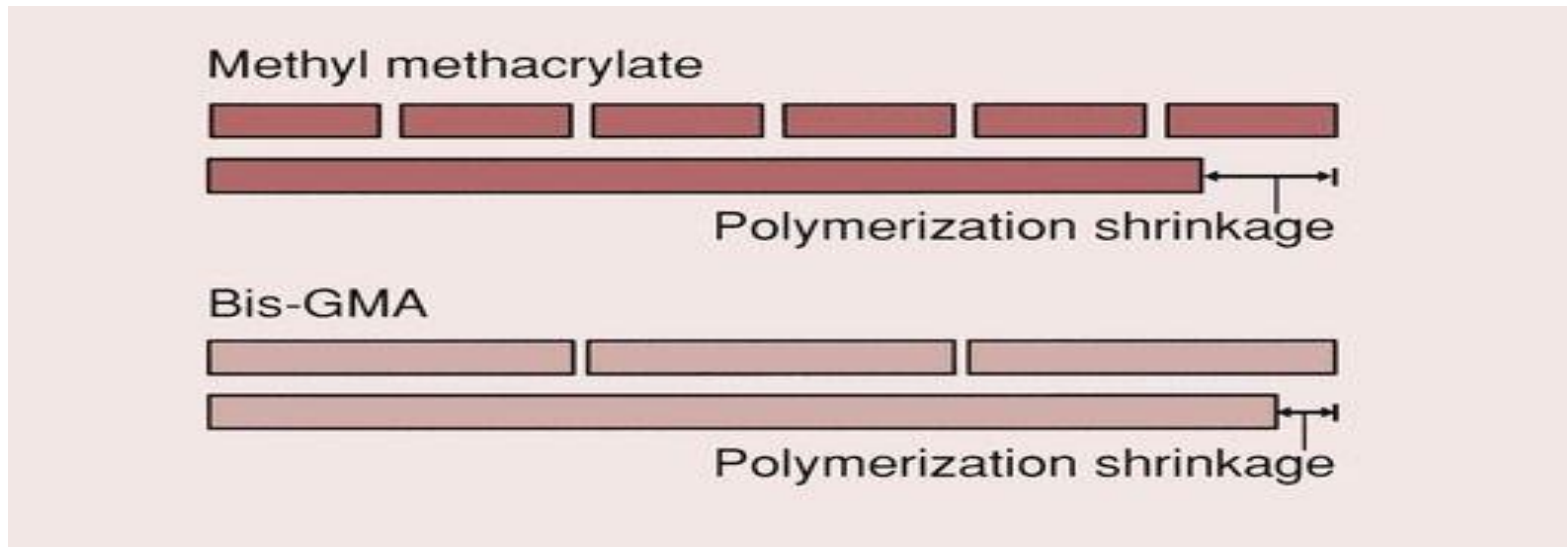
1. Responsible for hardening of the composite by polymerization reaction.
2. The dispersion medium that surrounds the inorganic fillers.
3. Bonds chemically with the adhesive bonding systems.

1. Organic matrix (Dispersion Medium):

a) High molecular weight monomer (Oligomer):

Advantages of Oligomers over methyl methacrylate monomers:

1- Lower polymerization shrinkage (due to higher molecular weight).



1. Organic matrix (Dispersion Medium):

a) High molecular weight monomer (Oligomer):

Advantages of Oligomers over methyl methacrylate monomers:

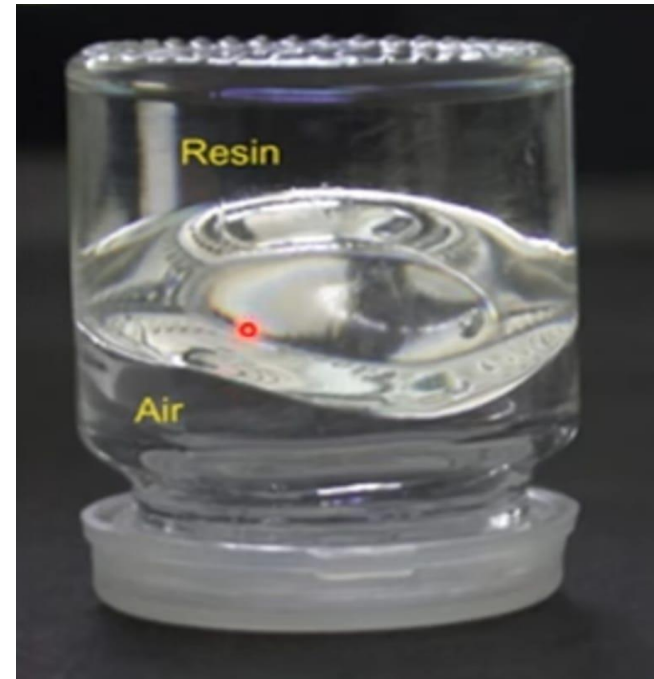
2. Lower volatility.
3. More rapid hardening.
4. Higher mechanical properties (due to higher molecular weight and higher cross linking).

1. Organic matrix (Dispersion Medium):

a) High molecular weight monomer (Oligomer):

Disadvantages of Oligomers:

High viscosity that limits the incorporation of inorganic fillers.



1. Organic matrix (Dispersion Medium):

b) Low molecular weight monomer (Diluent):

Types:

1. Diethylene glycol dimethacrylate (DEGDMA).
2. Triethylene glycol dimethacrylate (TEGDMA).

Function:

1. Reduce the viscosity of the matrix (allow incorporation of more fillers).
2. Facilitate the clinical manipulation.

Disadvantages:

1. Increase polymerization shrinkage.
2. Increase water sorption.

2. Inorganic fillers (Dispersed Phase):

Role of Fillers:

1. Improve the mechanical properties.
2. Reduce the polymerization shrinkage.
3. Decrease the coefficient of thermal expansion and contraction.
4. Reduce the heat evolved during polymerization .
5. Decrease water sorption.
6. Glass fillers reflect the color of the surrounding tooth structure so increase aesthetics.
7. Barium , strontium and zirconia fillers provide radio-opacity.

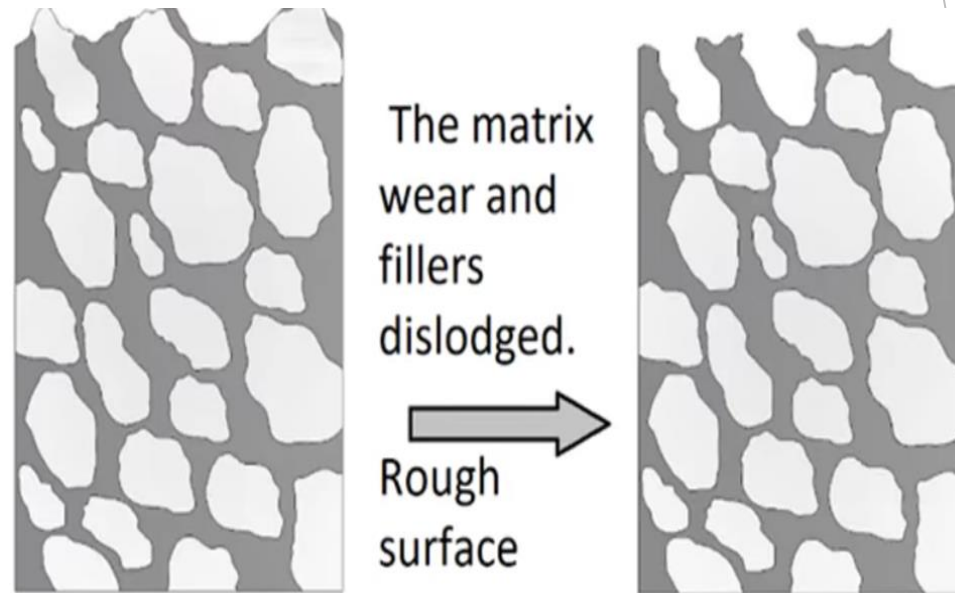
2. Inorganic fillers (Dispersed Phase):

Fillers size:

1. Macro-filled (conventional).
2. Micro-filled.
3. Micro-hybrid.
4. Nano-hybrid.
5. Nano-filled.

Fillers size:

- ▶ Composites with large filler size produce rough surface.
- ▶ Decreasing the filler size produce smoother surface after polishing but the finer particles have higher surface area.
- ▶ This high surface area leads to lower filler addition to matrix.

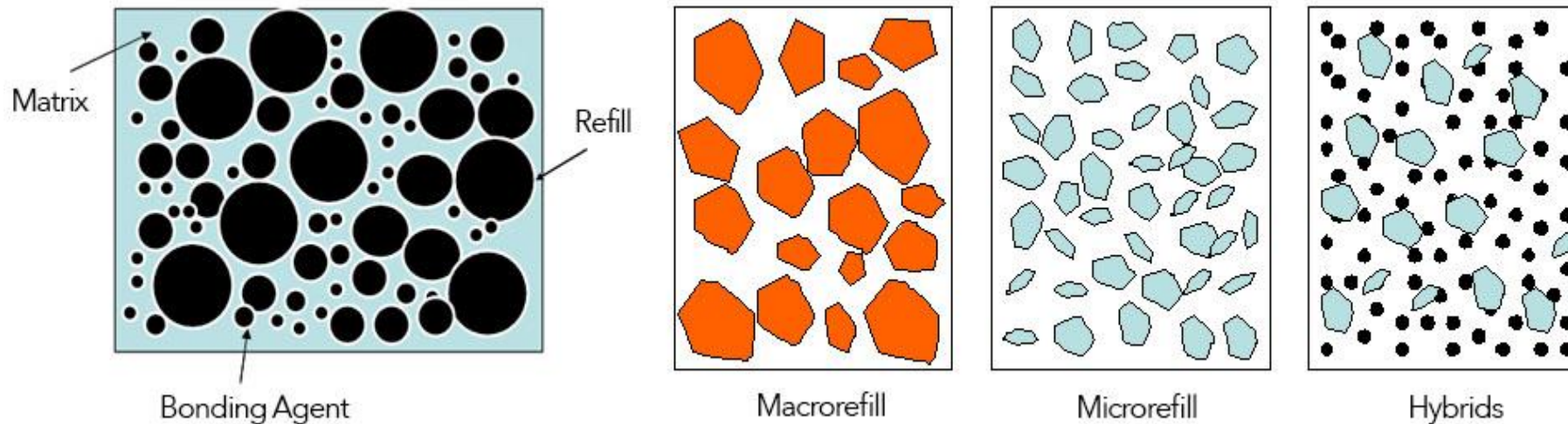


2. Inorganic fillers (Dispersed Phase):

Fillers Size:

↑ Filler size → ↑ % of Fillers, ↓ Aesthetics & ↑ mechanical properties.

↑ matrix → ↑ polymerization shrinkage, ↑ coefficient of thermal expansion, ↑ water sorption



2. Inorganic fillers (Dispersed Phase):

	Particle size (µm)	% Fillers (by Wt)	Advantages	Disadvantages
Conventional (Macro-filled composite)	20-30	78%	(Historical) • Good mechanical properties. • Irregular particles have better retention to the resin matrix.	• Difficulty in finishing & polishing. • Irregular surface due to erosion & loss of filler particles causing discoloration.

2. Inorganic fillers (Dispersed Phase):

	Particle size (μm)	% Fillers (by wt)	Advantages	Disadvantages
Fine	0.5-3	70-86%	• Better mechanical properties. • Better finishing & polishing.	• Aesthetics is not perfect.
Micro-fine	0.04-0.2	50-60%	• Easy finishing & polishing. • Good aesthetic (particles smaller than wavelength of light so give lustrous surface on polishing).	• Poor mechanical properties. • High polymerization shrinkage. (difficult to add large quantities of fillers as they have high surface area acting as thickening agent).

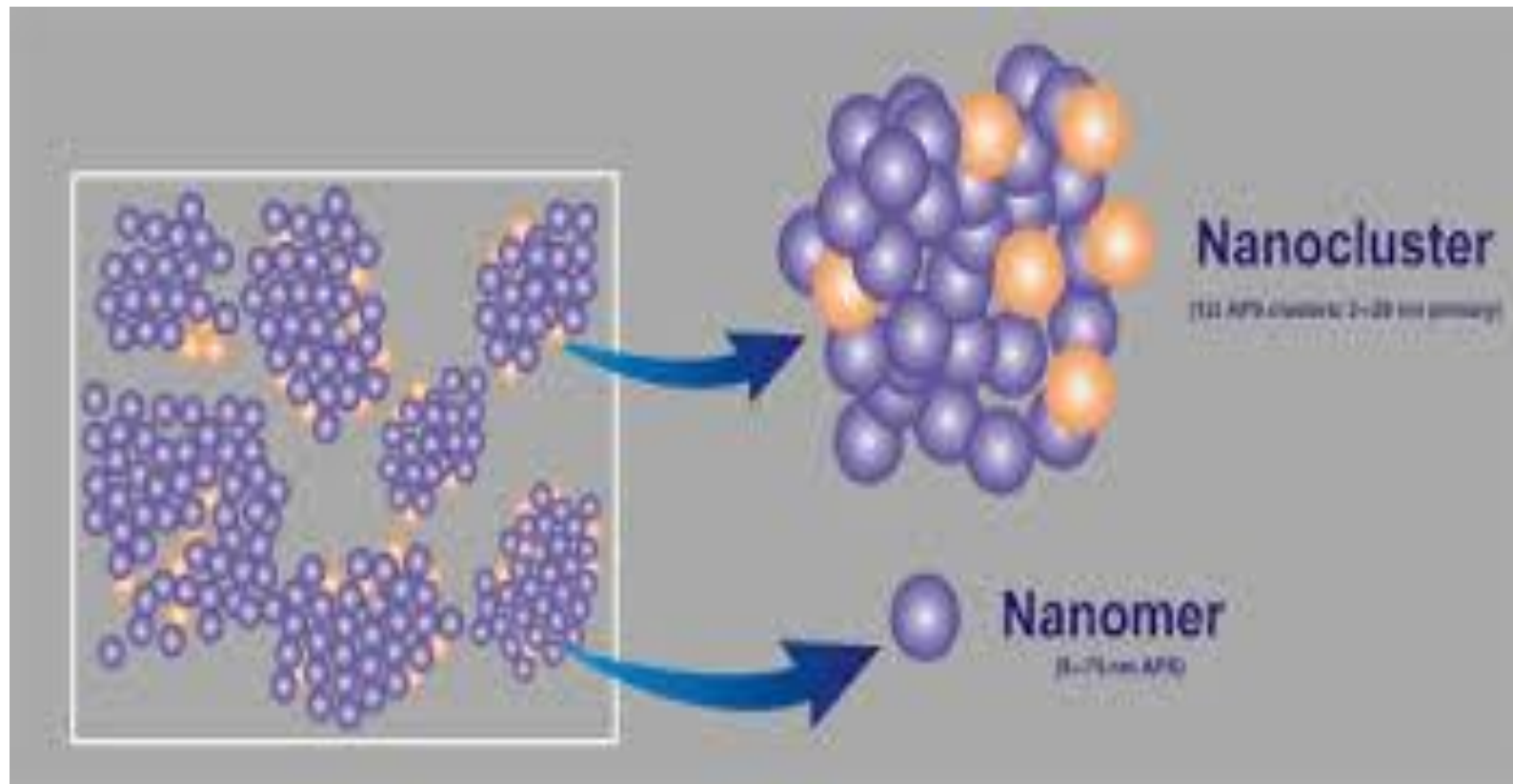
2. Inorganic fillers (Dispersed Phase):

	Particle size (μm)	% Fillers (by Wt)	Advantages	Disadvantages
Hybrid	Contain both Fine & Microfine fillers	Up to 70 % microfine	High mechanical properties (from fine fillers). Good surface finish(from microfine fillers).	
Nano-hybrid	Large particles (0.4-5) Nanosized particles			Not true nano filled resin composite.

2. Inorganic fillers (Dispersed Phase):

	Particle size (μm)	% Fillers (by Wt)	Advantages
Nano-filled	• Nanomers (spherical silica nanoparticles 2-75 nm). • Nanoclusters (spheroidal agglomerated particles 5-20 nm of silica & zirconia sintered together).	78%	• High mechanical properties (high filler content). • High translucency (filler size smaller than wavelength of light). • Reduced polymerization shrinkage. • Excellent polish ability and smoothness.

2. Inorganic fillers (Dispersed Phase):

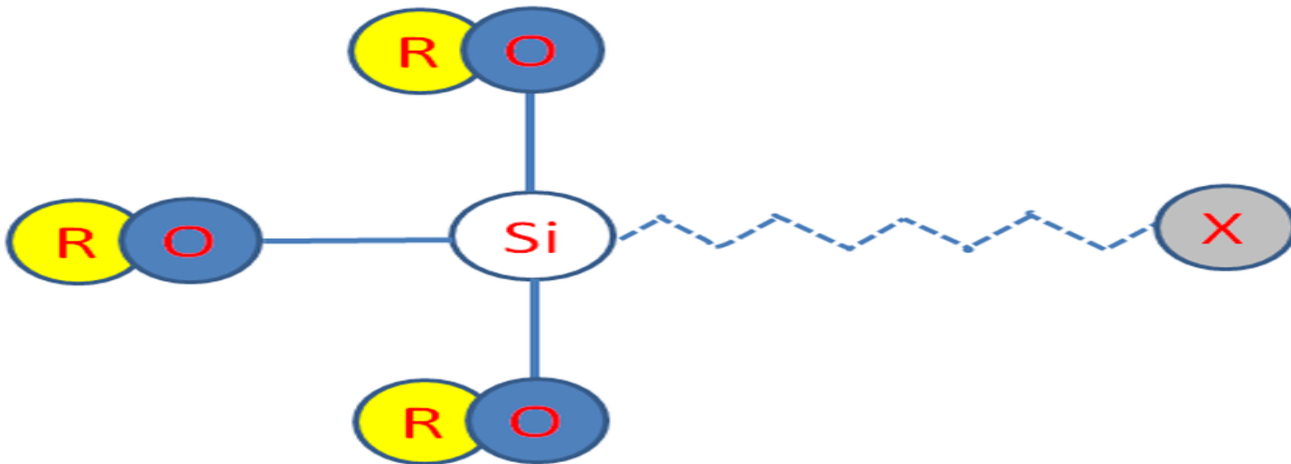


3- Coupling agents:

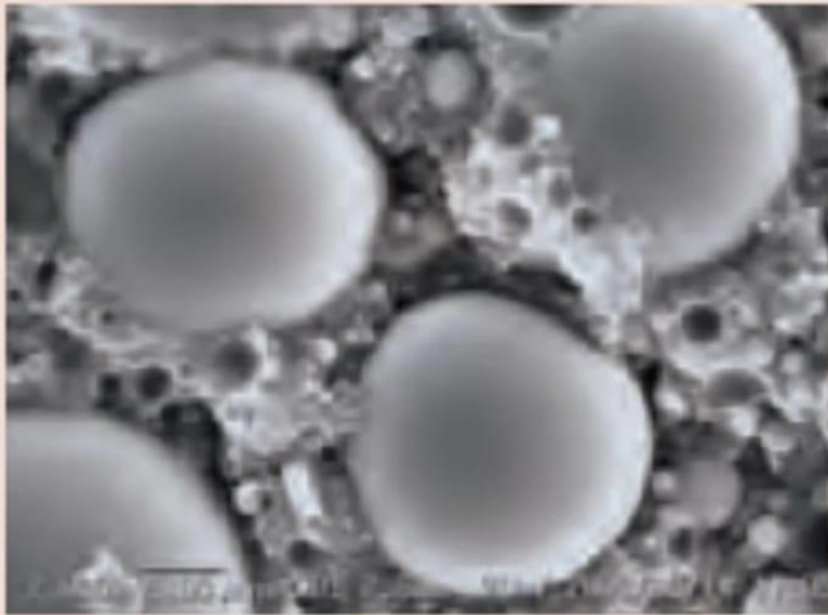
The usually used coupling agent is **vinyl silane**

It is molecule with two functional groups:

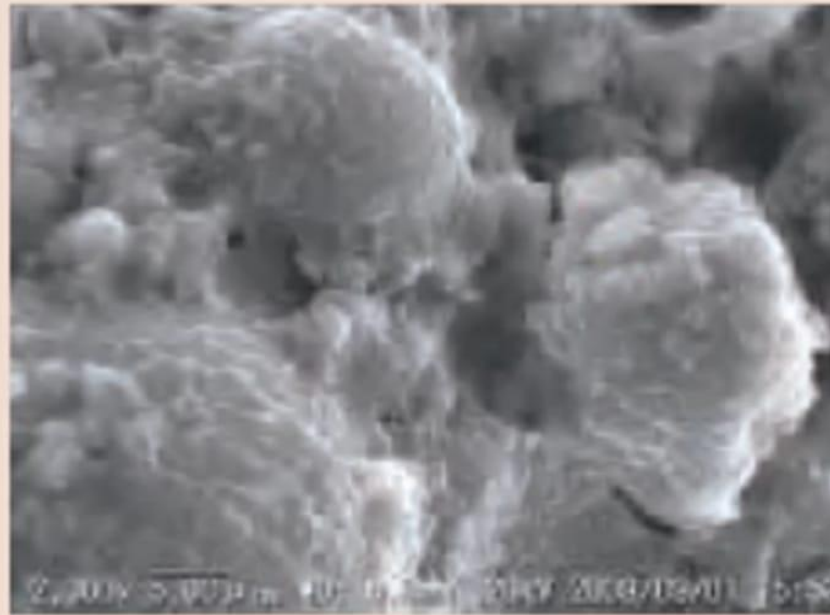
- ▶ Organic group bonds with matrix
- ▶ Inorganic group bonds with fillers



3- Coupling agents:



Untreated with silane coupling agents



Treated with silane coupling agents

3- Coupling agents:

Functions:

Binds the organic matrix with the inorganic fillers in order to:

1. Transfer the stresses from the weak matrix to the stronger filler improving the mechanical properties.
2. Prevent water penetration between matrix & fillers increasing hydrolytic stability.
3. Reduce early loss of fillers during service.

4. Initiator -Activator system

Polymerization reaction of resin composite is activated by:

1. Chemical activation.
2. Light activation.
3. Dual cure.

4. Initiator -Activator system

1. Chemical Cured Composite resin:

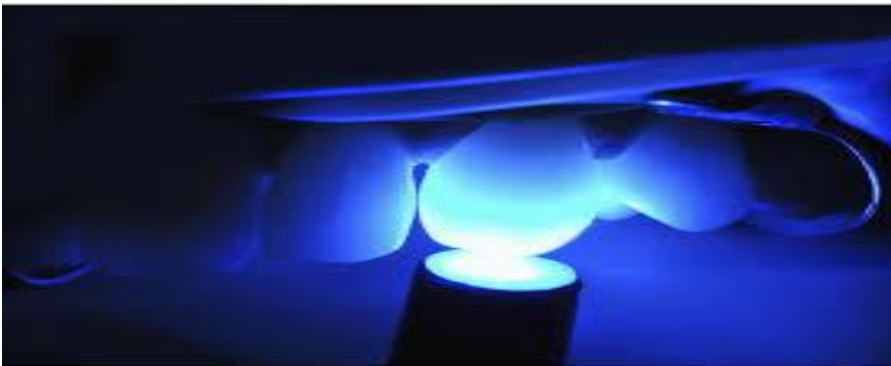
- ▶ It is supplied as 2 paste to be mixed:
 - a) One paste contains initiator (benzoyl peroxide).
 - b) The other contains activator (tertiary amine).



4. Initiator -Activator system

2. Light Cured Composite resin:

- ▶ It is supplied as single paste that requires no mixing.
- ▶ Blue visible light (wavelength ≈ 470 nm) activates the polymerization reaction.
- ▶ Initiator system is a mixture of **diketone** (as **camphorquinone**) and amine.



4. Initiator -Activator system

Advantages of Light Cured Composite over Chemical Cured types

1. No human variations as there is no mixing
2. No air bubbles.
3. Command setting time.
4. More color stability.
5. Less polymerization shrinkage (incrementally cured)

3. Dual Cured Composite resin:

- Contains initiator & accelerators that allow light activation followed by self curing.

5. Polymerization Inhibitors

- ▶ Inhibitors such as **hydroquinone** are added to the monomers to inhibit the premature polymerization reaction of the monomers during storage or accidental exposure to light.
- ▶ **Eugenol** inhibits polymerization of resin . So bases containing eugenol should not be used as base under composite.



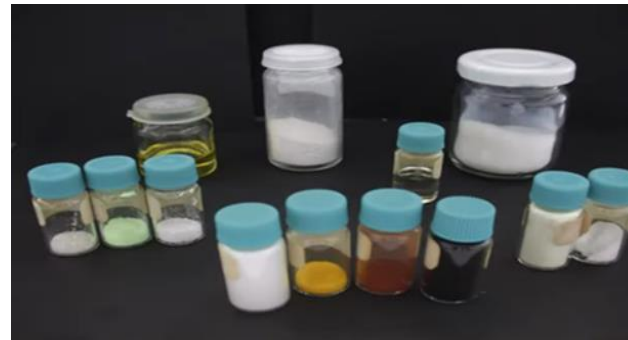
6. Ultraviolet Stabilizers

- ▶ Added to prevent discoloration (Yellowish) with time of resin composite restorations as they adsorb ultraviolet radiation.



7. Pigments

- ▶ Inorganic pigments are added to give the different shades of resin composite.



Properties of Resin Composite

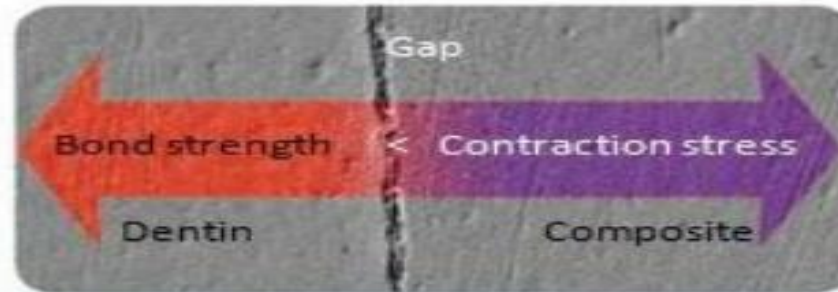
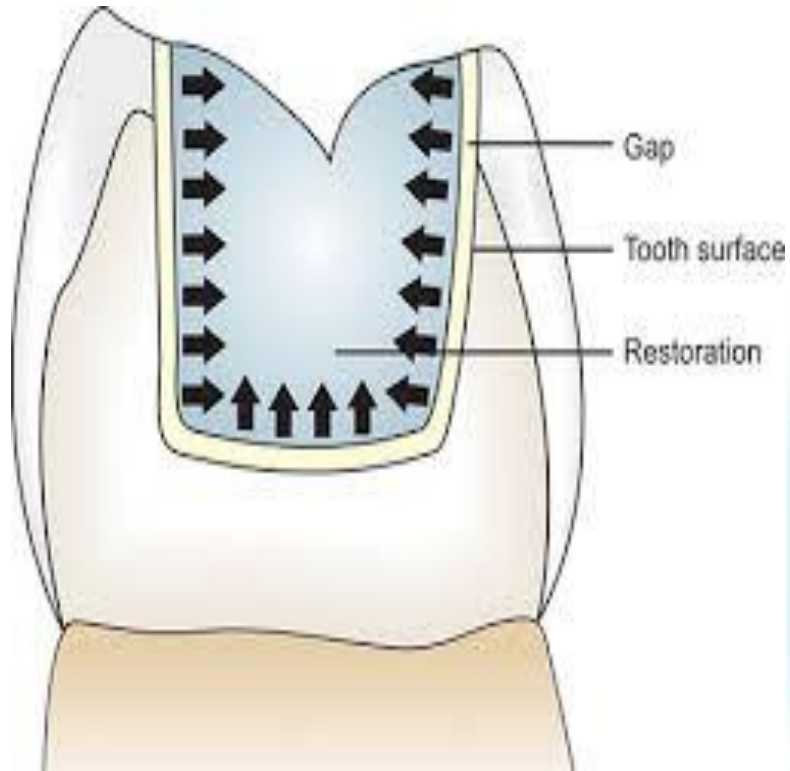
1. Polymerization shrinkage.
2. Bonding to tooth structure.
3. Thermal properties.
4. Mechanical properties.
5. Aesthetics and optical properties.
6. Solubility.
7. Water sorption.
8. Biological properties.

1. Polymerization shrinkage

- ▶ The matrix of resin composite undergo shrinkage during its polymerization which causes stresses inside the restoration.
- ▶ The polymerization increased as the amount of matrix increased.
- ▶ So microfine shrink more than hybrid and nano-filled types.
- ▶ If the polymerization stresses are higher than bond strength :

Bond failure → Gap formation → Microleakage → Recurrent caries and hypersensitivity

1. Polymerization shrinkage

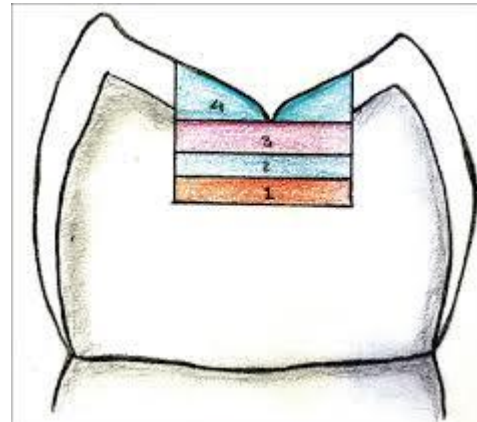


1. Polymerization shrinkage

Methods of decreasing polymerization shrinkage:

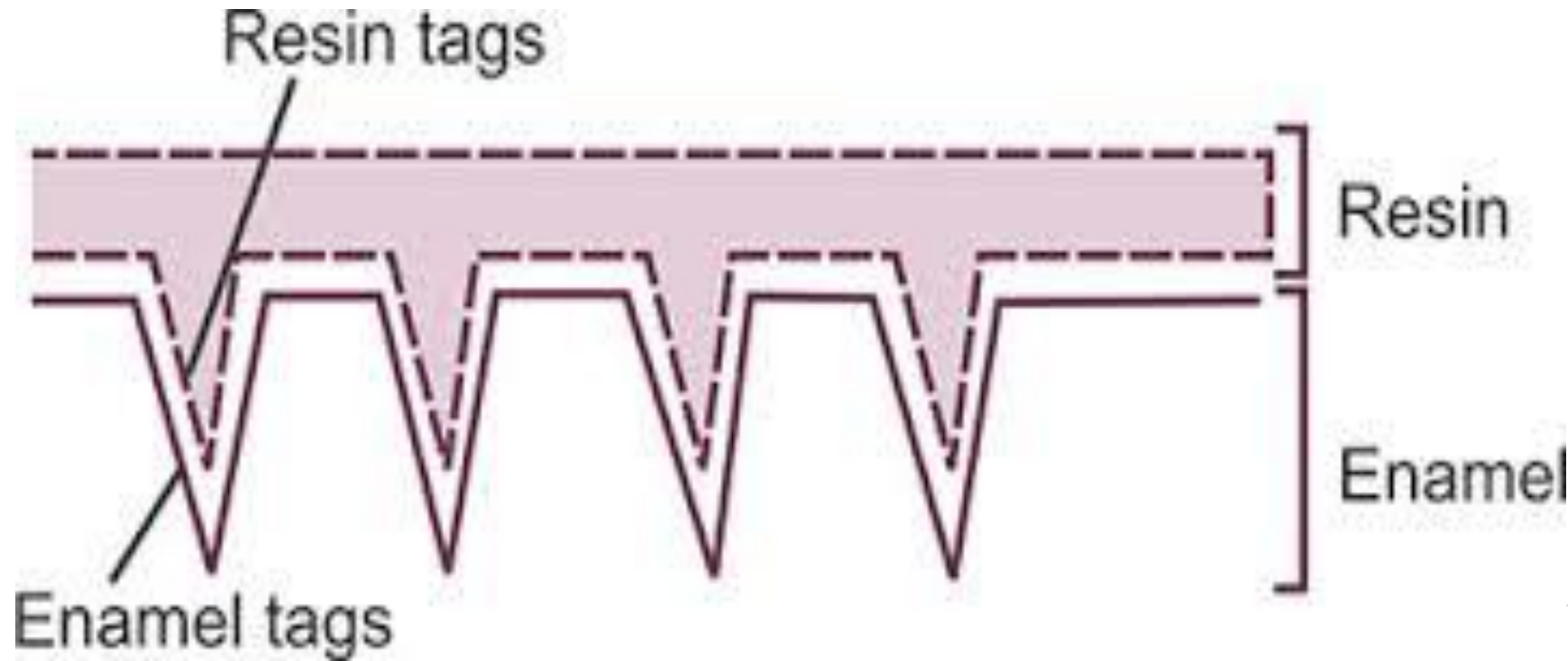
1. Incremental addition of light cured resin composite and polymerization of each increment independently.
2. Strong bond with tooth structure.
3. Indirect composite restoration:

The restoration is cured outside the mouth then cemented by resin cement to tooth structure.



2-Bonding to tooth structure

- ▶ Resin composite bonds **micro mechanically** with tooth structure by acid etching technique.



3-Thermal properties

a) Coefficient of thermal expansion and contraction:

- ▶ The higher the amount of matrix , the higher the coefficient of thermal expansion (the greater the change in dimensions due to changes in oral temperatures) causing more marginal leakage).
- ▶ Therefore, micro-filled resin composite has higher coefficient than hybrid & nano-filled types.

b) Thermal conductivity:

- ▶ Thermal conductivity of composites closely matches those of enamel & dentin.
- ▶ Therefore they provide good thermal insulation for dental pulp.

4- Mechanical properties

- ▶ Increasing the fillers amount improves the mechanical properties.
- ▶ Micro-filled composite has inferior mechanical properties compared with hybrid and nano filled types.

5- Aesthetics and optical properties

- ▶ Resin composite is a translucent material that can match tooth color.
- ▶ It is supplied in different shades for more simulation of natural teeth.
- ▶ Nano-filled composite has high translucency as the fillers size is smaller than light wavelength.

6- Solubility

- ▶ It has lower solubility.
- ▶ The solubility resulting from release of residual monomers.

7- Water Sorption

- ▶ Resin matrix is responsible for water sorption .
- ▶ Micro-filled composite showed more water sorption than hybrid and nano-filled types.

8. Biological Properties

- ▶ Residual monomers & uncured deep layers may cause chemical irritation of the dental pulp.
- ▶ Rough surface of restoration may lead to plaque accumulation.

Low shrinkage Resin composite

1. Ormocer (Organically Modified Ceramics):

- ▶ The matrix is formed of organic- inorganic copolymer
- ▶ This copolymer is consisting of a siloxane backbone and an organic methacrylate side chains connected to the siloxane backbone.

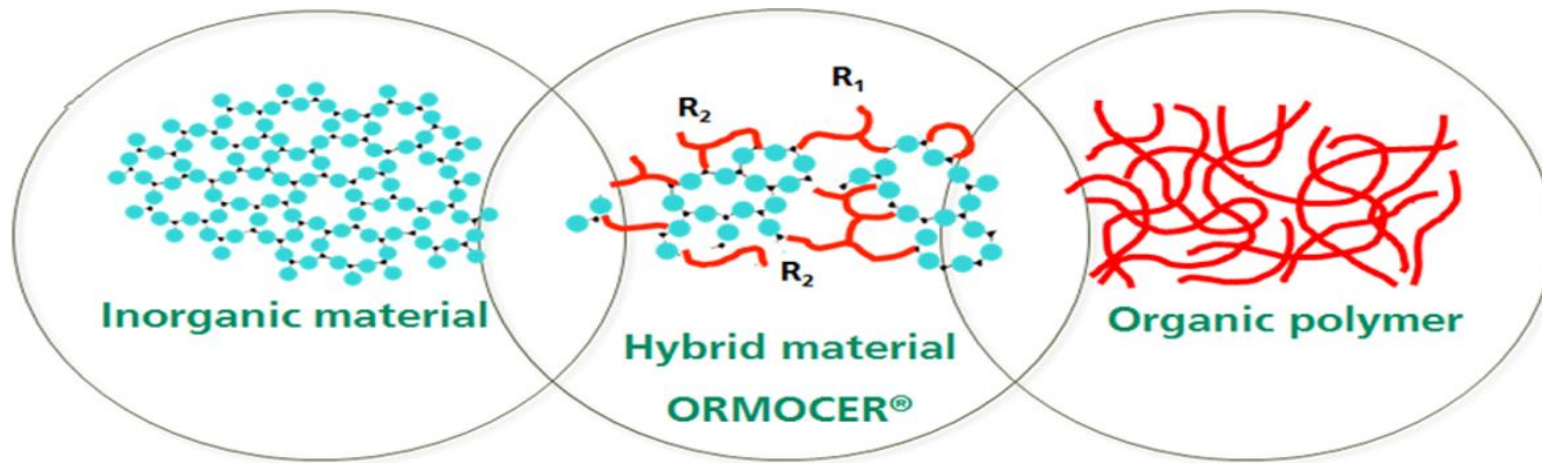


Figure 11 • Schéma matrice ORMOCER®10

Low shrinkage Resin composite

1. Ormocer (Organically Modified Ceramics):

- ▶ During curing, the organic methacrylate side chains undergo free radical addition polymerization.
- ▶ The siloxane molecule is longer than Bis-GMA molecule which is responsible for the lower polymerization shrinkage of ormocer.

Low shrinkage Resin composite

1. Ormocer (Organically Modified Ceramics):

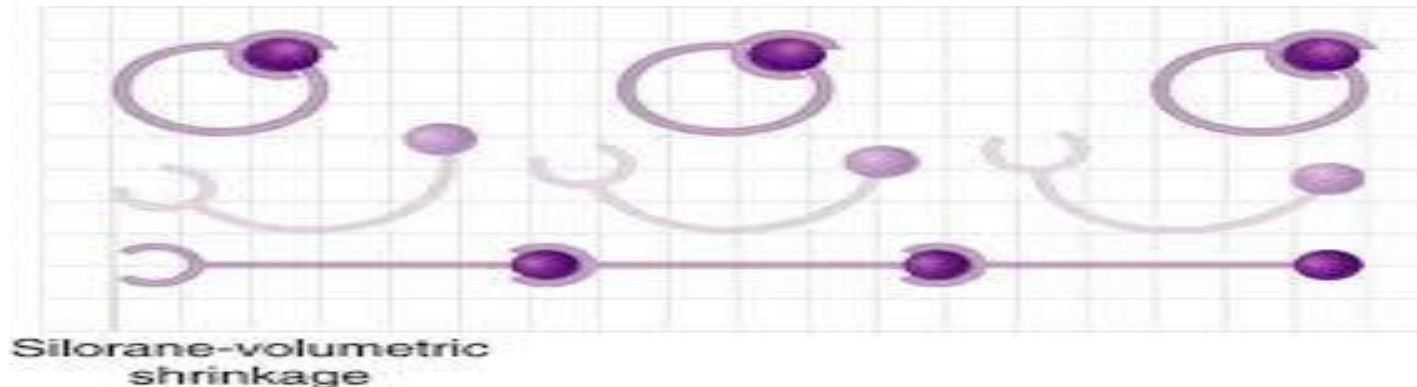
► Advantages of Ormocer:

1. Low polymerization shrinkage.
2. Highly biocompatible.
3. Good esthetics.
4. More depth of polymerization.

Low shrinkage Resin composite

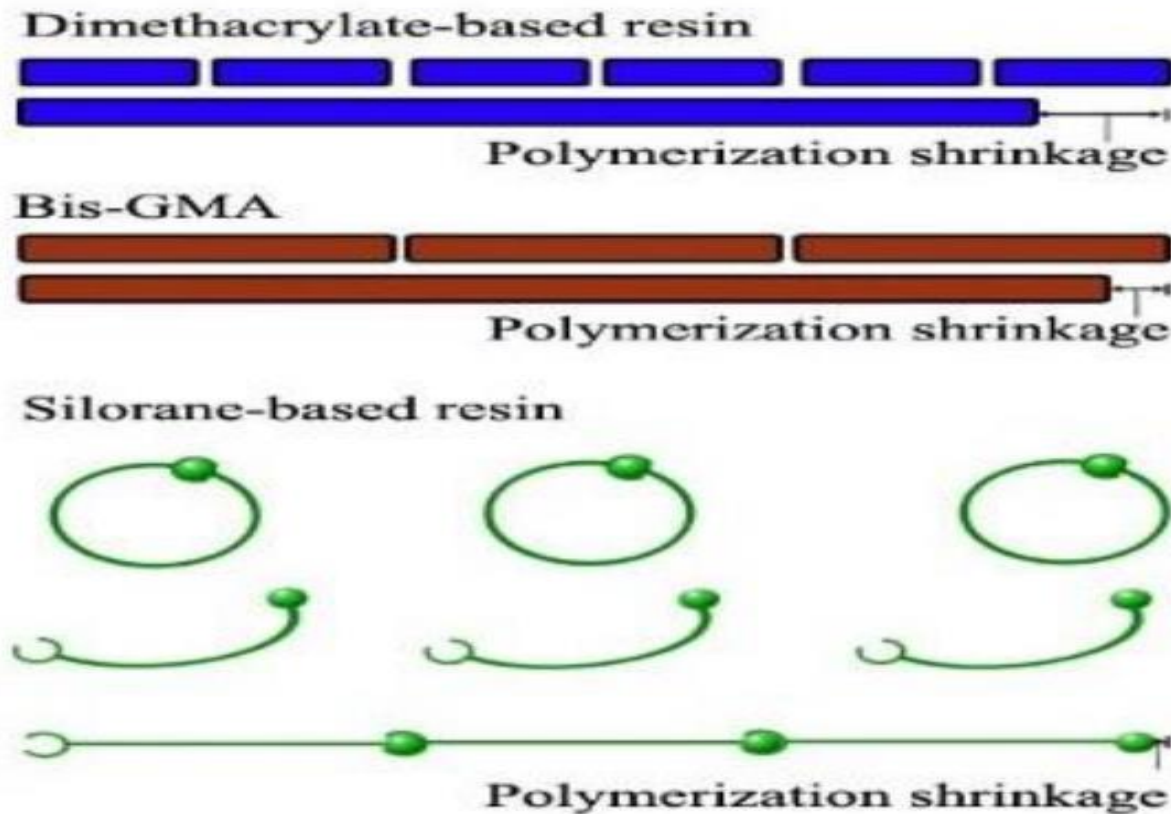
2.Silorane:

- ▶ The matrix is siloxane and oxyrane.
- ▶ The terminal group is ring instead of double bond.
- ▶ It is polymerized by ring opening polymerization reaction.
- ▶ Polymerization shrinkage less than 1%.



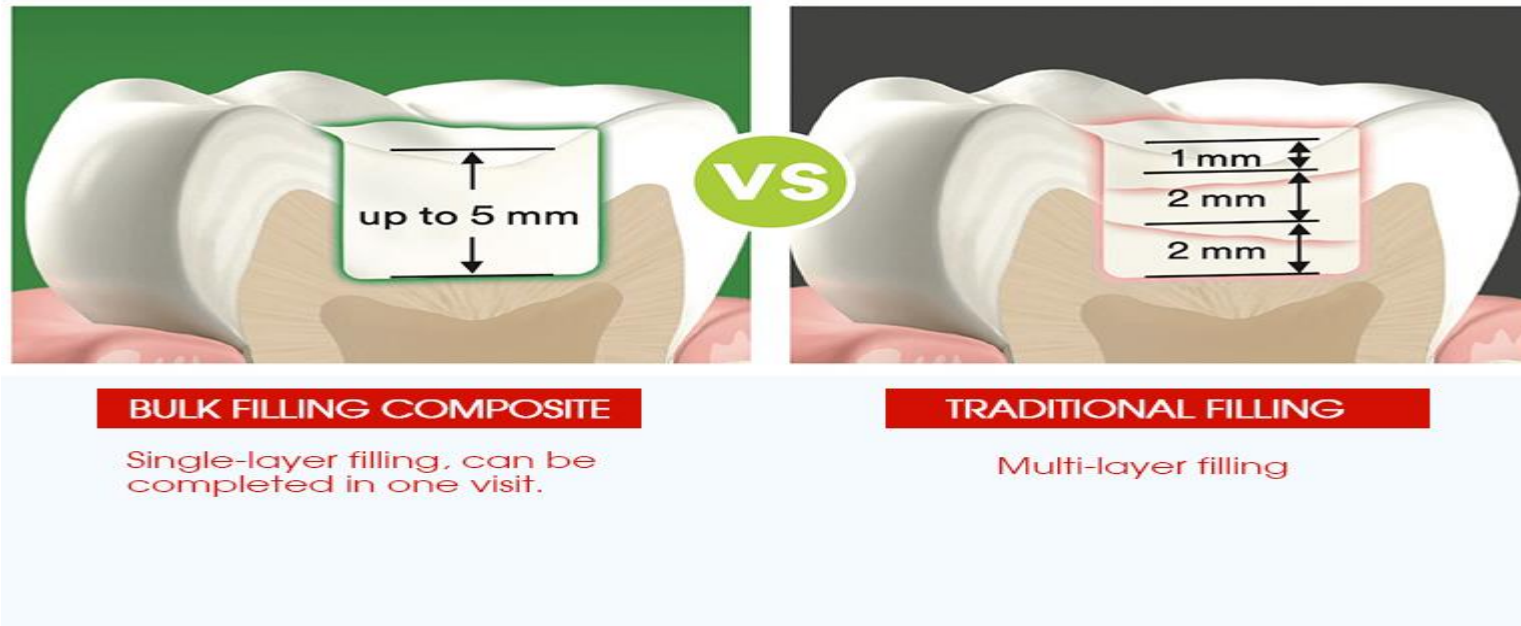
Low shrinkage Resin composite

2.Silorane:



Bulk -fill Composites

- ▶ This is a special type of composite that is able to build -up the restorations in bulky layers up to 4 mm in thickness.
- ▶ The effective depth of curing and polymerization stresses are the main problems facing this type.



Packable Resin Composite

- ▶ It is high viscosity resin composite.
- ▶ Used for posterior restorations and core build up.

Advantages:

1. Packing against matrix allowing good proximal contact formation.
2. Non-sticky.

Disadvantages:

1. Poor adaptation to the cavity wall due to high viscosity.
2. Opaquer than hybrid resin composite with comparable mechanical properties.

Flowable Resin Composite

- ▶ It is low viscosity composite.
- ▶ It is dispensed into cavity by syringe with needle tip.
- ▶ Used as a liner or for cervical lesion restorations.
- ▶ Low viscosity is achieved by decreasing filler content.



Flowable Resin Composite

► Advantages:

1. Better adaptation to the cavity wall due to lower viscosity.
2. Lower polymerization stresses due to lower elastic modulus (causes stress relaxation).

► Disadvantages:

1. Lower mechanical properties.
2. High polymerization shrinkage.

Classification of Resin Composite

1. According to particle size & shape:

- a) Fine Composite
- b) Micro-fine Composite
- c) Hybrid Composite
- d) Nano-hybrid Composite
- e) Nano-filled Composite

2. According to method of curing:

- a) Chemically cured composite
- b) Light cured composite
- c) Dual cured composite

3. According to viscosity:

- a) Conventional composite
- b) Packable Composite
- c) Flowable composite

Thank You